

Task 52: Top K Frequent Elements

Problem Description

Given an integer array `nums` and an integer `k`, return the `k` most frequent elements. You may return the answer in any order.

Java Solution

```
class Solution {
    public int[] topKFrequent(int[] nums, int k) {
        Map<Integer, Integer> countMap = new HashMap<>();
        for (int num : nums) {
            countMap.put(num, countMap.getOrDefault(num, 0) + 1);
        }

        PriorityQueue<Integer> pq = new PriorityQueue<>(
            (a, b) -> countMap.get(a) - countMap.get(b)
        );

        for (int key : countMap.keySet()) {
            pq.offer(key);
            if (pq.size() > k) {
                pq.poll();
            }
        }

        int[] res = new int[k];
        for (int i = k - 1; i >= 0; i--) {
            res[i] = pq.poll();
        }
        return res;
    }
}
```